Created using a free version of SMath Studio

$$y(x) := \sqrt{x}$$

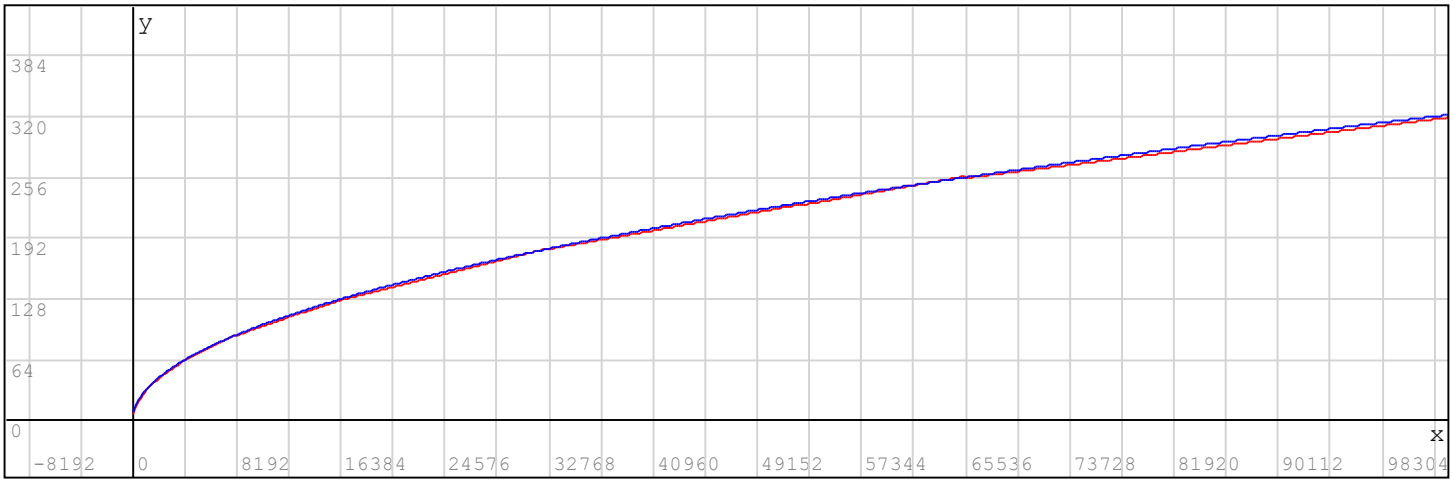
$$clz(x) := \begin{cases} 32 & \text{if } x = 0 \\ 32 - \operatorname{trunc}\left(\log_2\left(\operatorname{trunc}(x) + \frac{2^{32}-1}{2^{32}}\right) + \frac{2^{32}-1}{2^{32}}\right) & \text{else} \end{cases}$$

$$msb(x) := 32 - clz(x)$$

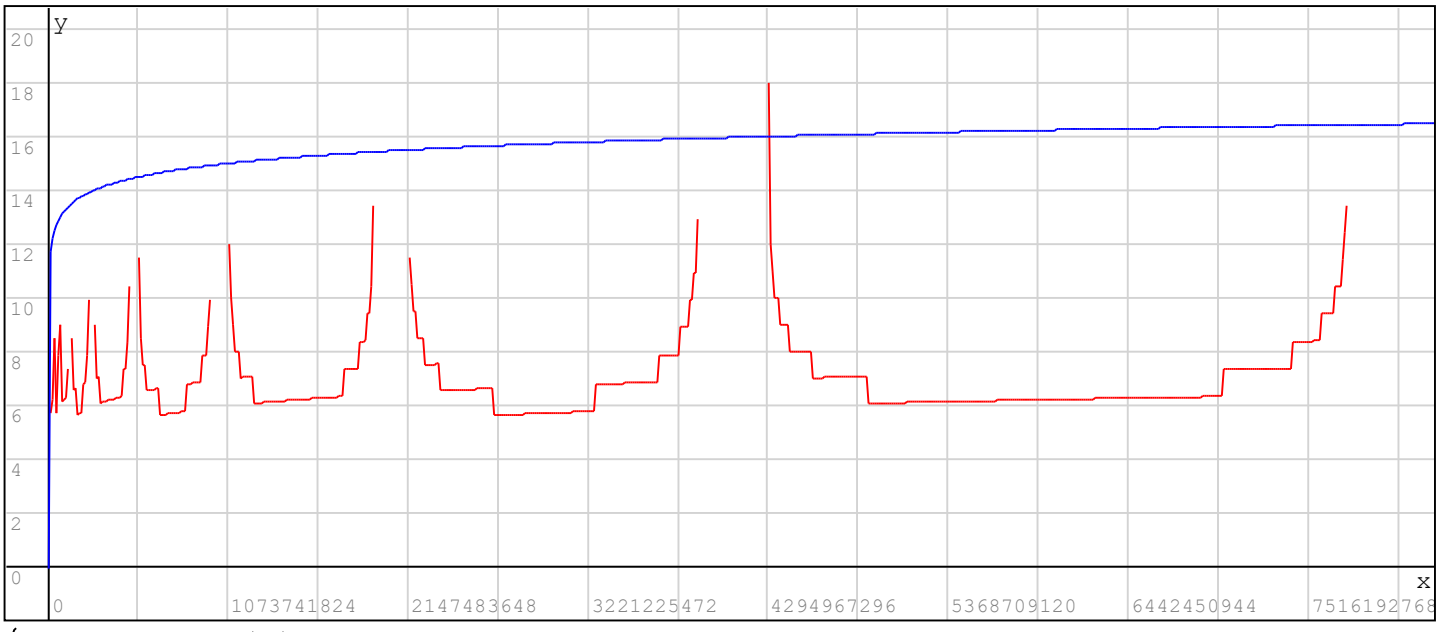
$$linSqrt(x) := \operatorname{trunc}\left(\frac{\left(\operatorname{trunc}(x) - 2^{msb(x)-1}\right)}{2^{msb(x)-1}} \cdot \operatorname{trunc}\left(\frac{\operatorname{trunc}\left(\left(\sqrt{2}\right)^{msb(x)}\right) \cdot \operatorname{trunc}\left(\left(1+\sqrt{2}\right)\cdot 2^{29}\right)}{2^{32}}\right)\right) + \operatorname{trunc}\left(\sqrt{2^{ms}}\right.$$

$$linearBitError(x) := \operatorname{trunc}\left(\log_2\left(\left(\sqrt{\operatorname{trunc}(x)} - linSqrt(x)\right)\right) + \frac{2^{32}-1}{2^{32}}\right)$$

$$linearBitValid(x) := \log_2\left(\sqrt{\operatorname{trunc}(x)} + \frac{2^{32}-1}{2^{32}}\right)$$



$$\begin{cases} \sqrt{x} \\ linSqrt(x) \end{cases}$$



$$\begin{cases} linearBitValid(x) \\ linearBitValid(x) - linearBitError(x) \end{cases}$$

Not for commercial use